

CS 550: Computer Graphics and Geometric Design

Course Code	CS 550
Course Name	Computer Graphics and Geometric Design
L-T-P-C	2-0-2-3
Prerequisites	IC 111 Linear Algebra or similar course, IC250 Data Structure and Algorithms or similar course.
Intended for	B.Tech. (CSE, EE and ME) II/III/IV year/ MS/M.Tech /PhD (any branch)
Distribution	Elective for B.Tech. II/III/IV year/ MS/M.Tech/PhD
One-time approval	37th BoA (Oct 2020); 30th Senate (Feb 2021), in the M.Tech Computer Science and Engineering curriculum

Course Contents

- **Introduction:** Motivation; Use of computer graphics in different domains; Digital display technologies; Color; Graphics processing pipeline; OpenGL. [2 Lectures]
- **Geometry representation:** Lines, triangles, polygons, meshes, point-clouds, polynomials, B-splines; Parametric v/s. implicit representation, boundary representation, volumetric representation. [2 Lectures]
- **Rasterization:** World and camera transformations; Orthogonal and perspective projections; Hidden-surface removal, Z-buffering; Bresenham's line drawing algorithm; Triangle drawing primitive; Flood-fill algorithm. [6 Lectures]
- **Lighting, shaders and texture mapping:** Radiometric concepts such as BRDF, BTDF, BSSRDF; Types of light sources such as point, spot, parallel; Texture mapping; Shading schemes such as Flat, Gouraud, Phong; Flattening of 3D surfaces; Bump maps, Normal maps, Displacement maps. [6 Lectures]
- **Ray tracing and ray casting:** Mechanism of tracing the path of light in order to generate realistic images from 3D scenes. [3 Lectures]
- **Modeling with curves:** Differential geometry of planar and space curves; Definition and basic properties of Bezier and B-spline curves; Weierstrass approximation theorem. [5 Lectures]
- **Modeling with surfaces:** Differential geometry of surfaces; Definition and basic properties of tensor-product Bezier surfaces. [4 Lectures]

Laboratory Modules

Lab to be conducted on a 2-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course shall be appropriately sequenced for synchronization with the laboratory.

- Lab 1: Installation of Blender, familiarization of the GUI, simple exercises involving creation of primitive shapes. Installation of OpenGL, writing a 'Hello World' application. Draw a few parametric as well as implicitly defined curves.
- Lab 2-4: Implementation of Bresenham's line drawing algorithm. Implementation of flood fill algorithm.
- Lab 5-7: Create a 3D scene in Blender and apply various models of light-interaction such as BRDF, BTDF, BSSRDF. Insert different types of light sources into the scene. Apply textures to 3D objects. Lighting and shading in OpenGL, Raytracing
- Lab 8-10: Implementation of ray-tracing.
- Lab 11-12: Installation of Open Cascade. Construct and plot Bezier curves. Plot the tangents and osculating circles at a few points. Plot the Frenet frame at a few sampled points. Construct and plot B-spline curves.
- Lab 13-14: Construct and plot tensor-product Bezier surfaces. Plot tangent-plane and curvature at sampled points. Extract iso-curves.

Text Books

1. Donald D. Hearn et al., **Computer Graphics with OpenGL**, 3rd Edition, Pearson, 2013

References

1. Steve Marschner et al., **Fundamentals of Computer Graphics**, 4th Edition, A K Peters/CRC Press, 2015
2. Gerald Farin, **Curves and Surfaces for CAGD**, 5th Edition, 2001
3. Dave Shreiner, **OpenGL Programming Guide: The Official Guide to Learning OpenGL**, Versions 3.0 and 3.1 (7th Edition)